CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT334 - G01

PDCA DOCUMENT
BLOCKNOTES

Members:

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I. PLAN

In this phase, identify the aspects of your **existing app** you want to improve, or outline your **course of action** if you are developing a proposed app.

Feature 1: User Authentication and Security

- **Current Problem:**
  The application completely lacked a user authentication system. There was no login or sign-up functionality. All notes were public and visible to any visitor, and there was no way to attribute notes to a specific author. This presented a major security flaw and made it impossible to provide a personalized user experience.

- **Proposed Solution/ Plan for Implementation:**
  The plan is to implement a robust and secure user authentication system using a trusted third-party provider to streamline the login process.

1. Select Authentication Tool: Integrate the django-allauth library, a comprehensive and secure package for handling social authentication in Django.
2. Choose Provider: Use Google as the sole OAuth provider for its ubiquity and ease of use.
3. Configure Django Project:
   - Add django-allauth and its dependencies (sites, socialaccount, etc.) to INSTALLED_APPS in settings.py.
   - Configure the necessary AUTHENTICATION_BACKENDS and set the SITE_ID.
   - Update the project's main urls.py to include the allauth URL patterns under the /accounts/ path.
4. Create Google OAuth Credentials:
   - Set up a new project in the Google Cloud Console
   - Configure the OAuth consent screen.
   - Create an "OAuth 2.0 Client ID" for a Web application, ensuring the authorized redirect URI is set correctly to http://127.0.0.1:8000/accounts/google/login/callback/.
5. Secure Credentials:
   - Create a superuser account for the Django admin panel.

- Store the Google Client ID and Client Secret securely by creating a "Social Application" object within the Django admin interface, linking it to the correct site. This is the production-ready method for managing credentials.

6. Protect Views: Secure the main notes view by adding the @login_required decorator to ensure only authenticated users can access it.

Feature 2: User Interface and Experience (UI/UX)

- **Current Problem:**
  The application's user interface was highly minimalistic and unstyled. It consisted of a single, hardcoded dark theme with no option for user preference, which could be an accessibility issue. The layout was basic, and key interactive elements were either missing or not intuitive, leading to a poor user experience.

- **Proposed Solution/ Plan for Implementation:**
  The plan is to implement a complete visual and interactive overhaul of the application based on a high-fidelity Figma prototype, transforming it into a polished and immersive Web3 experience.

  1. Adopt New Color Palette & Background:

     * Implement the new dark, moody color palette (#1A092D, #32255A, etc.

  2. Implement Glassmorphism:

     * Rebuild key UI components (sidebar, modals) as "glassmorphic" cards using the backdrop-filter: blur() CSS property.

  3. Redesign Core Pages:

     * Landing Page: Re-create notes/landing.html to match the new design with the centered layout and glassmorphic feature cards.

     * Notes Interface: Overhaul /notes.html to implement the new two-column layout with a glassmorphic sidebar.

     * Logout Page: Redesign logout.html to use the glassmorphic confirmation modal.

4. Enhance Interactivity and Animations:

   * Add a cyan glow effect to the "Blockchain Online" status indicator.

   * Implement a real-time polling mechanism in JavaScript to check the Ganache connection every 5 seconds and update the status indicator's style and pulsing dot animation.

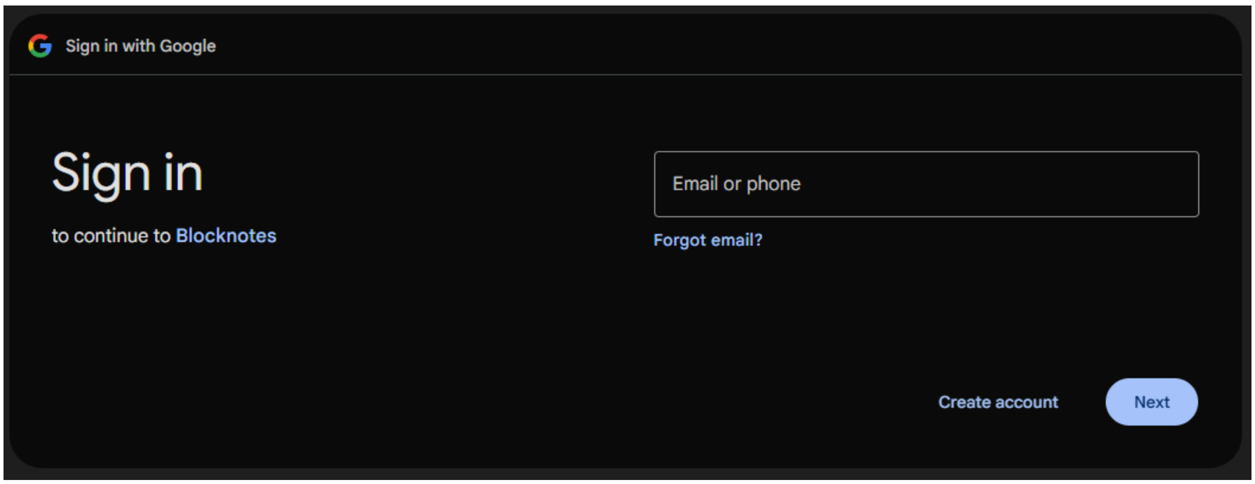
   * Apply gradient text effects and neon glows to key headings and buttons.

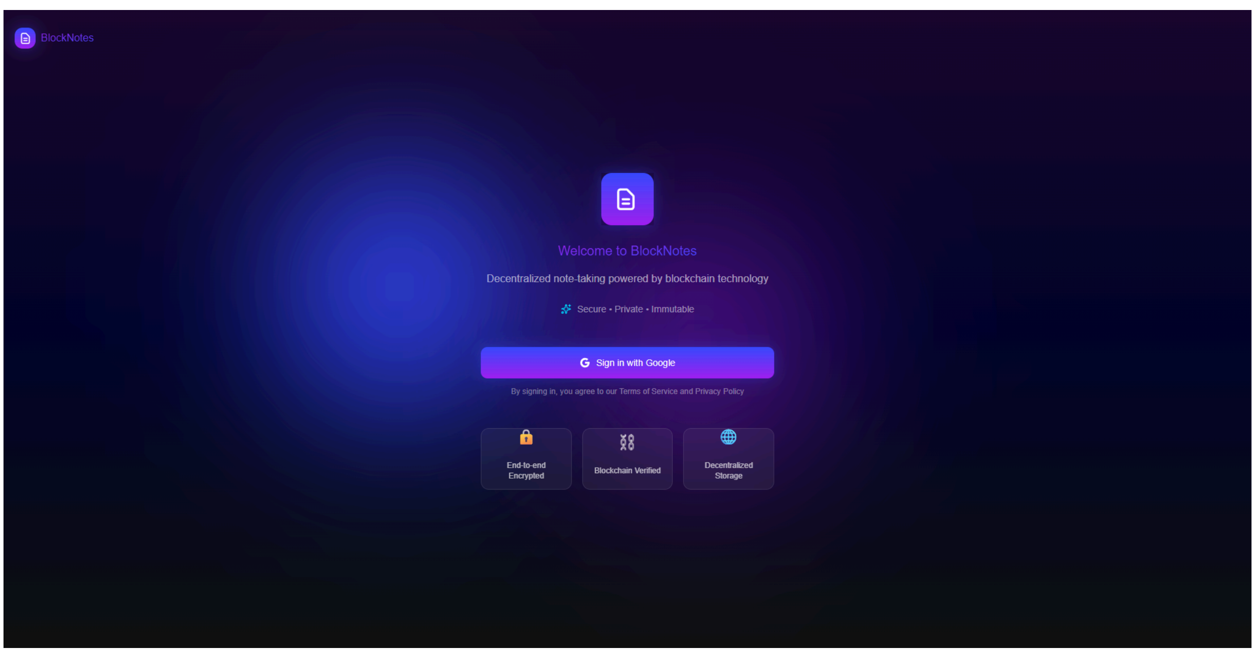
5. Enhance Transaction Receipt Panel:
   *Added a dedicated proof page for notes that are save on blockchain.

II. DO

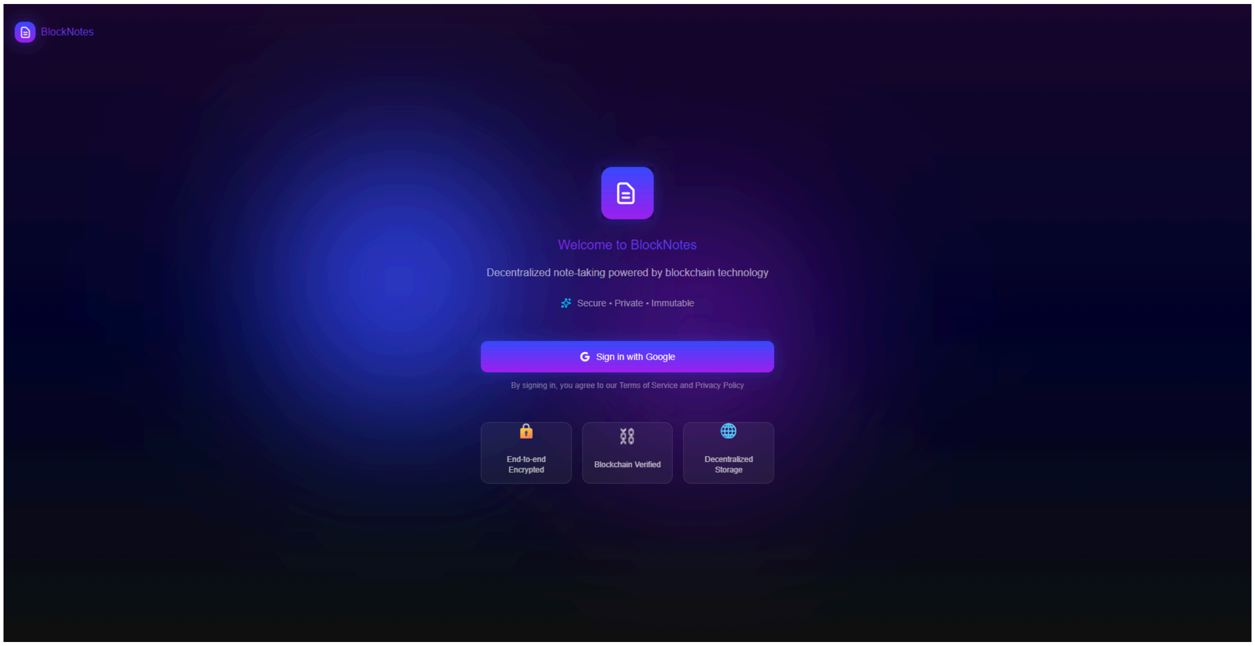
In this phase, provide **evidence of implementation** for the planned features. Include **screenshots or images** of your work showing at least **80% completion** for each feature.

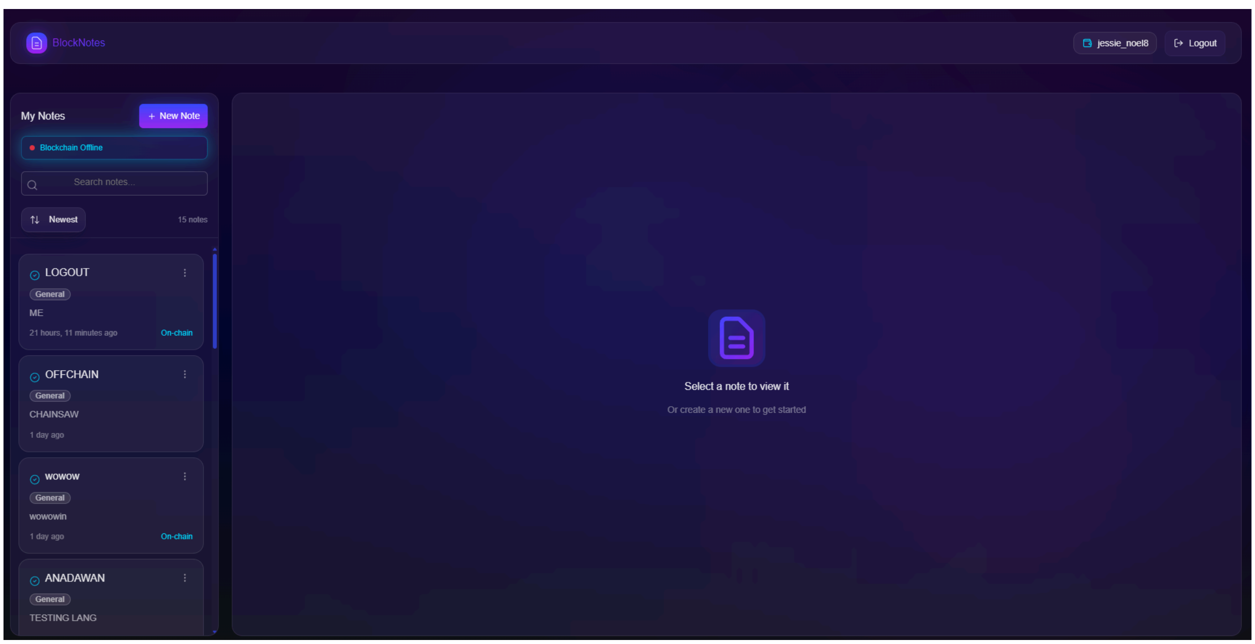
Feature 1: User Authentication and Security

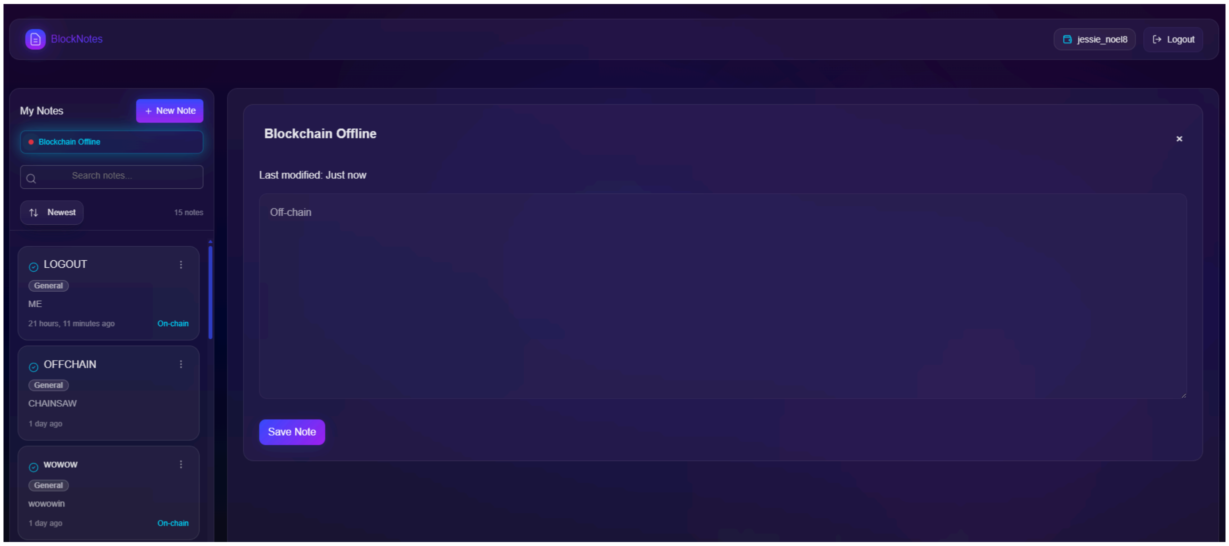


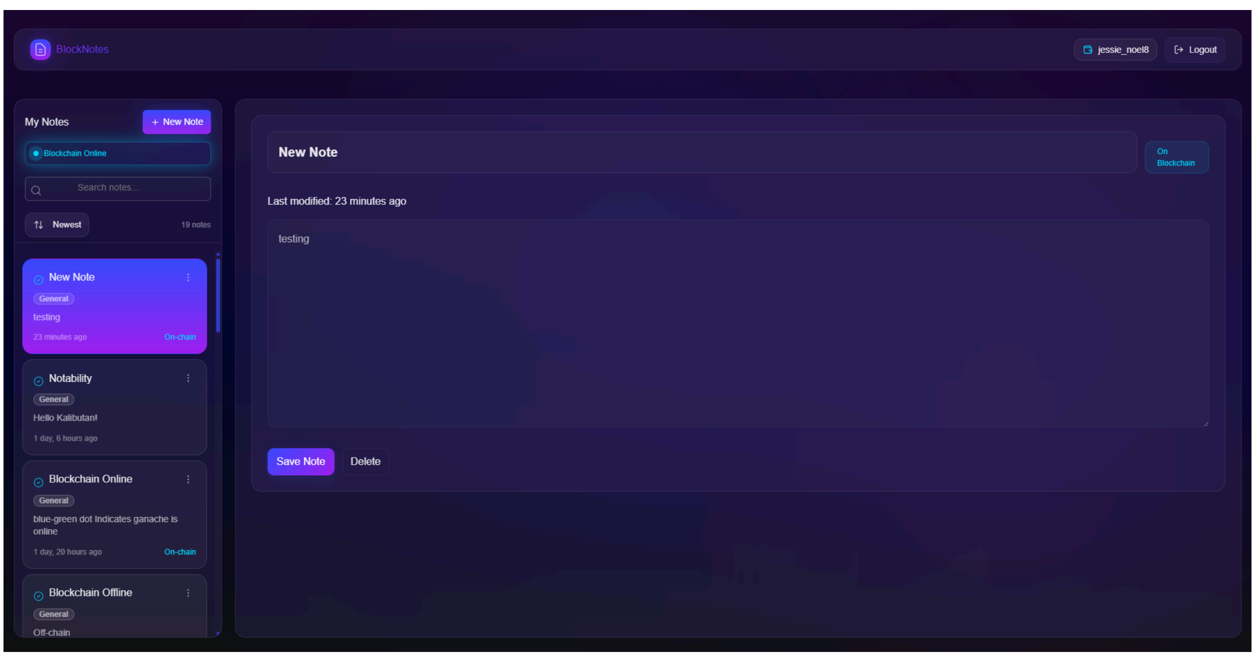


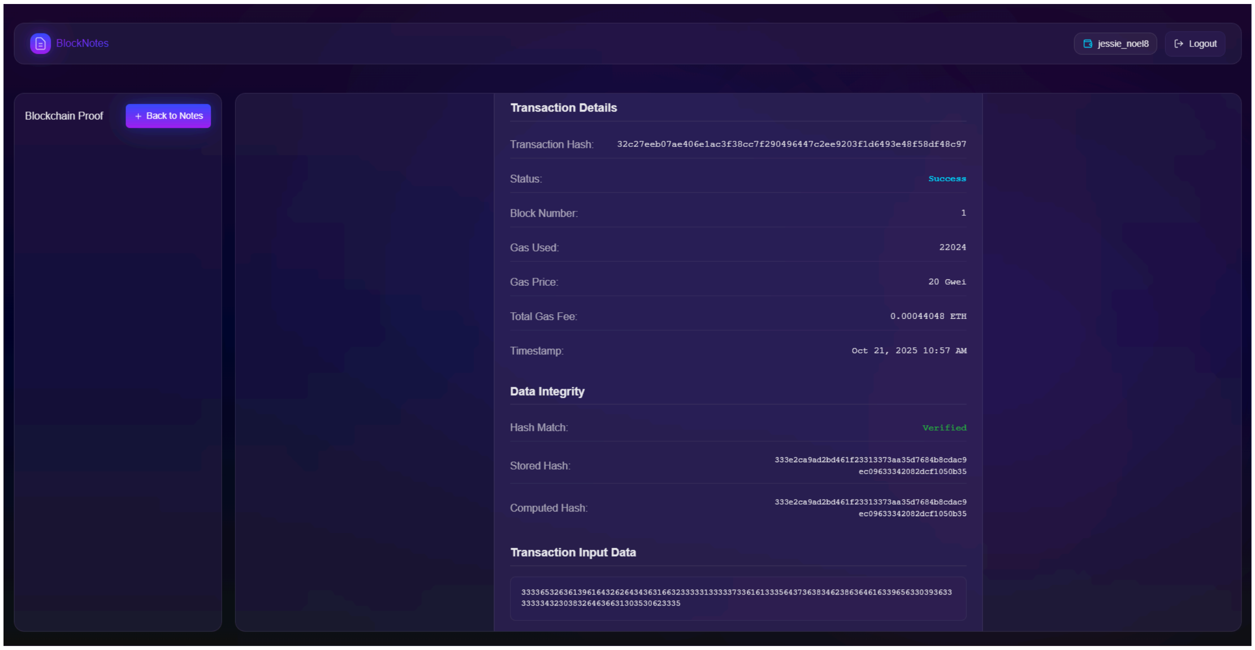
Feature 2: User Interface and Experience (UI/UX)






BlockNotes
jessie_noel8
Logout
My Notes
New Note
Blockchain Offline
Search notes...
Newest
15 notes
LOGOUT
General
ME
21 hours, 11 minutes ago
On-chain
OFFCHAIN
General
CHAINSAW
1 day ago
wowow
General
wowowin
1 day ago
On-chain
Blockchain Offline
Last modified: Just now
Off-chain
Save Note


BlockNotes
jessie_noel8
Logout
My Notes
New Note
Blockchain Online
Search notes...
Newest
19 notes
New Note
General
testing
23 minutes ago
On-chain
Notability
General
Hello Kalibutan!
1 day, 6 hours ago
Blockchain Online
General
blue-green dot indicates ganache is online
1 day, 20 hours ago
On-chain
Blockchain Offline
General
Off-chain
New Note
On Blockchain
Last modified: 23 minutes ago
testing
Save Note
Delete


BlockNotes
jessie_noel8
Logout
Blockchain Proof
+ Back to Notes
Transaction Details
Transaction Hash: 32c27eeb07ae406e1ac3f38cc7f290496447c2ee9203f1d6493e48f58df48c97
Status: Success
Block Number: 1
Gas Used: 22024
Gas Price: 20 Gwei
Total Gas Fee: 0.00044048 ETH
Timestamp: Oct 21, 2025 10:57 AM
Data Integrity
Hash Match: Verified
Stored Hash: 333e2ca9ad2bd461f23313373aa35d7684b8cdac9ec09633342082dcf1050b35
Computed Hash: 333e2ca9ad2bd461f23313373aa35d7684b8cdac9ec09633342082dcf1050b35
Transaction Input Data
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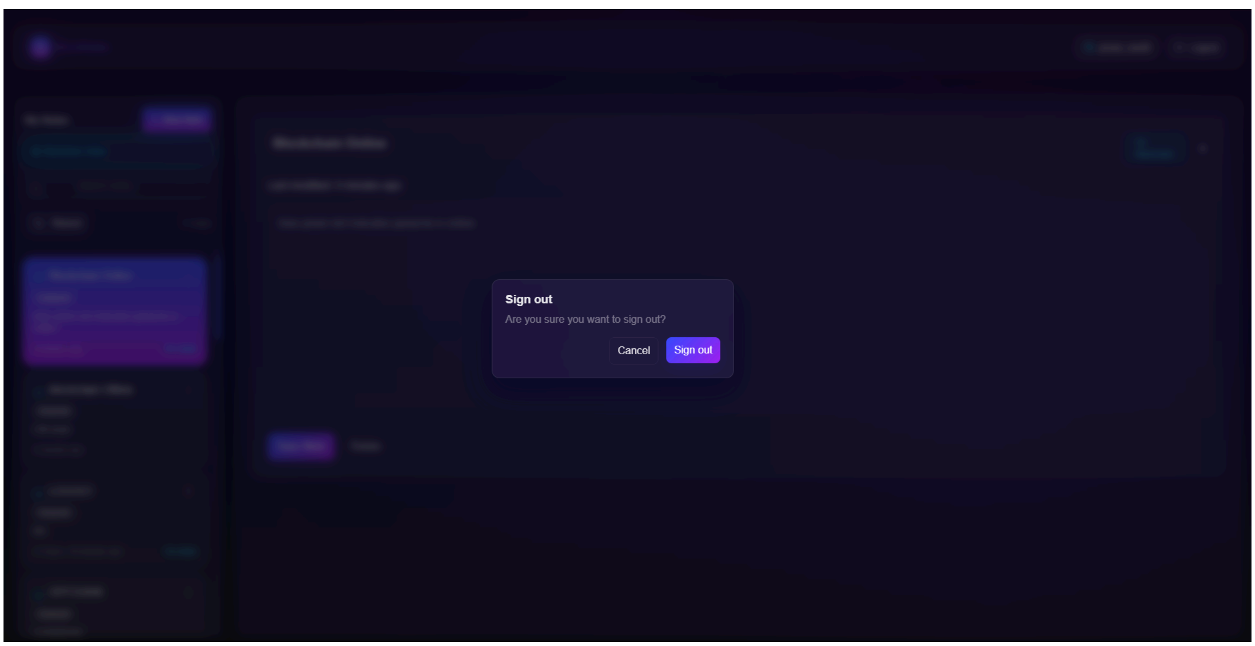
333334323038326463663130353062335


Sign out
Are you sure you want to sign out?
Cancel
Sign out